

MobileNet – ConvLSTM-Based Low-Power Spatio - Temporal Surveillance for Border Security

**A PROJECT REPORT SUBMITTED IN PARTIAL
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AWARD OF THE DEGREE OF**

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IN**

COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

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We, the undersigned, declare that the Major Project titled “**MobileNet-ConvLSTM -Based Low-Power Spatio -Temporal Surveillance for Border Security**” has been carried out by us in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech.)** in **COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE** of Jawaharlal Nehru Technological University Hyderabad (JNTUH), under VNRVJIET B.Tech. R22 Academic Regulations, at Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology, Hyderabad.

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ABSTRACT

Most existing border surveillance systems rely mainly on spatial information extracted from individual video frames, which limits their ability to recognize motion-based infiltration activities. To overcome this limitation, this work employs a MobileNet–ConvLSTM model that learns both spatial, temporal characteristics from spying video. MobileNet takes spatial features from each frame, and ConvLSTM learns temporal patterns from a series of frames that come one after the other. This lets the system correctly sort different types of infiltration events. The complete MobileNet–ConvLSTM model is deployed on a single-board computer connected to a camera for real-time video capture. When an infiltration event is detected, the relevant data is being sent to a central server, where it is kept in a database for further analysis and monitoring. To train the model, The UCF Crime Dataset and VisDrone VID dataset are used through which the model’s ability is increased for different scenarios. The proposed model achieves an accuracy of 92.80% during training while maintaining low computational complexity. Due to its lightweight design, the system can run efficiently on low-powered edge devices, making it suitable for deployment in remote border areas where power and computational resources are limited.

Keywords: Border Surveillance, MobileNet-ConvLSTM, Spatio-Temporal Feature Learning, Infiltration Detection, Real-Time Video Monitoring.

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NOMENCLATURE

CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
ConvLSTM	Convolutional Long Short-Term Memory
AI	Artificial Intelligence
ML	Machine Learning
YOLO	You Only Look Once
YOLO-X	Extended version of YOLO object detection model
YOLO-R	YOLO-based model variant (used for detection tasks)
SIFT	Scale-Invariant Feature Transform
UCF	University of Central Florida (UCF-Crime Dataset)
VID	Video Object Detection (VisDrone-VID Dataset)
GPU	Graphics Processing Unit
TPU	Tensor Processing Unit
CPU	Central Processing Unit
NIPS	Neural Information Processing Systems (Conference)
GPU	Graphics Processing Unit
TPU	Tensor Processing Unit
CPU	Central Processing Unit
RNN	Recurrent Neural Network

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CHAPTER 1

INTRODUCTION

1.1 Background Of the Problem

Monitoring borders in isolated and resource-limited areas has historically depended on high-resolution cameras and human oversight. Although recent developments in Deep Learning have greatly enhanced automated surveillance functions, the majority of current systems evaluate video footage on a frame-by-frame basis. This method emphasizes spatial data while neglecting temporal dynamics, which are crucial for comprehending motion patterns and human actions. Consequently, these systems face challenges in identifying intricate infiltration activities such as stealthy movements or coordinated intrusions, resulting in diminished accuracy and a rise in false alarms.

1.2 Need and Relevance

There is an urgent need for intelligence-based border surveillance systems that would operate efficiently in conditions of severe limitations on available power, storage space, and computing resources. Border locations, more often remote and dangerous sites, are not supported by stable infrastructure including powerful servers, reliable power supply, or broadband Internet. Therefore, applying traditional approaches to training deep learning models, which are energy-hungry and require GPUs, becomes difficult. This limitation necessitates a shift toward lightweight and energy-efficient architectures that can operate directly on edge devices without relying on cloud-based processing. Edge computing will not only reduces latency but also improves the reliability and security by minimizing dependence on external communication networks, which may be unavailable or vulnerable in border areas. Therefore, models was designed for such applications can strike a careful balance between computational efficiency and prediction performance.

Temporal structure will plays an equally important role in improving surveillance systems. The problem with traditional frame-based systems is that they are unable to detect motion dynamics, which was an integral part of any intelligent surveillance system to detect activities like crawling, loitering, or synchronized group behaviour.

Using temporal models in the analysis of consecutive frames using methods like ConvLSTM enables this capability.

Additionally, fast spatial features extraction techniques, made possible by models like MobileNet, ensure that the algorithm is able to remain accurate while lowering computational load considerably. Both of these capabilities like spatial intelligence and temporal intelligence are making it feasible for the algorithm to create strong surveillance systems.

In this regard, the creation of an efficient, portable, and smart surveillance system can prove to be extremely advantageous. These types of systems have the ability to provide reliable and timely detection even under the most difficult conditions, thus fulfilling an important need in the realm of surveillance technology.

1.3 Problem Statement

There are quite a few issues associated with the use of contemporary border surveillance technologies when it comes to identification of infiltration behaviour. First of all, the issue in question relates to the technique utilized by the systems in order to analyse the data contained in the video footage. To be more exact, one should admit that the analysed information is processed on its own without any regard to its context. This means that these systems focus on spatial aspects of analysis of videos including sizes, shapes, and appearances of detected objects only.

Furthermore, there have been recent developments in Deep Learning that have made it possible to develop very accurate models that can be used to analyse videos. Nevertheless, such models are often developed to work in high-end computer systems, and they feature complex neural networks that have a lot of parameters. These models require a great deal of computation power, memory, and energy. For these reasons, they cannot be deployed in border areas due to their high resource requirements.

Thus, the main issue lies in two aspects: one is that the existing approaches do not have the capability to use temporal knowledge to comprehend behavior; and the other is that it is difficult to implement the complex algorithms on devices with limited computing

resources. To solve this issue, we need to design an efficient and robust intrusion detection system that uses temporal knowledge directly on edge devices.

1.4 Objectives

Objectives of this project are to:

1. Develop a hybrid light-weight framework which can perform both spatial and temporal features extraction.
2. Use the MobileNet architecture to perform spatial features extraction in an efficient manner through depth wise separable convolutional operation.
3. Implement the ConvLSTM layer for extracting temporal dependency between consecutive frames in the video.
4. Deploy the trained architecture to perform surveillance on a low-power edge computing device such as the Raspberry Pi 5.
5. Train the developed model through multi-stages of transfer learning by using VisDrone-VID and UCF Crime datasets.

1.5 Scope of Work

This paper focuses on the process behind the design and implementation of a lightweight spatio-temporal surveillance model, which can be used to conduct surveillance at the borders in real-time. The model will be designed and trained using suitable data sets, and implemented using an embedded system with an onboard processor. Importance is placed on the balance between speed and accuracy. The paper is limited to testing using predetermined data sets only.

In this research, the goal is to create and use an efficient spatiotemporal surveillance system that can be used for border surveillance. The scope of work will cover the creation of the model, training the model on appropriate datasets, and implementation on a single-board computer. While emphasis will be placed on balancing performance with accuracy, limitations of the study include working only with datasets and test environments.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of Existing Work

The development of intelligent video surveillance system has been significantly along with the improvements using deep learning methods, specifically in the field of automatic pattern recognition from visual sources. Traditional video surveillance systems will heavily relied on traditional spatial patterns and rules, which were found insufficient for detecting complex behaviour-based anomalies. It was only through the advantage of Convolutional Neural Networks (CNN) that has intelligent video surveillance became capable of accurately recognizing spatial info from single frames in video formats. LeCun et al. [1] and Krizhevsky et al. [2] provided the theoretical basis for deep learning based on CNN, which is now the fundamental framework of current visual recognition technologies. Simonyan and Zisserman [5] successfully showed how two-stream CNN networks can be used for action recognition tasks and served as a milestone in the development of spatial information extraction from videos.

While the spatial modeling abilities of CNNs remain highly effective, there is a crucial drawback that exists in these models – their incapability of modeling any temporal dynamics across successive frames. In other words, as CNNs treat each video frame independently of one another, they do not model the trajectory of movement, behavioral change, and time-dependent relationships that are required for detecting any anomaly in a surveillance video stream. To solve this problem, a number of researchers have looked into the possibility of integrating CNNs with RNNs, particularly with LSTM networks developed by Hochreiter and Schmidhuber [6]. The ability of LSTM networks to store sequential data for extended periods of time makes them ideal candidates for detecting activities like walking, running, climbing, or loitering.

Nevertheless, there exists a significant issue in the usage of conventional LSTMs in video applications because of the need to flatten the spatial feature map into one-dimensional vectors during input, which leads to the loss of spatial information. This flattening of the feature maps breaks down the spatial dependencies within the feature map, making traditional LSTM networks incapable of spatially related tasks like intrusion detection. In order to tackle this problem, Shi et al. [9] proposed the concept of Convolutional Long Short-Term Memory Networks, which involves replacing the

fully connected transformation within the LSTM cells with convolutional layers. This allows the network to maintain spatial information during the entire temporal learning process, enabling modeling of both the identity and motion of an object simultaneously.

The domain of anomaly detection using surveillance is well supported by works like Sultani et al. [13], which proposed the UCF-Crime data set and proposed an approach for real-world anomaly detection using weak supervision techniques. Luo et al. [10] has suggested sparse coding techniques which was within the stacked RNN architectures to enable unsupervised intrusion detection. Chalapathy and Chawla [11] was offered an overview of the application of deep learning models has able to anomaly detection, discussing both supervised and unsupervised models for the same. However, although these models have helped in understanding how anomalies can be modelled, most of the models have been computationally expensive, requiring GPUs that cannot be used in restricted settings.

This problem of deploying such intelligent systems in edge-computing environments, especially in scenarios related to border surveillance, has pushed researchers towards lightweight and computationally efficient deep learning architectures. The MobileNet family, specifically the versions MobileNet and MobileNetV2 presented by Howard et al. [3] and Sandler et al. [4], respectively, employ depth-wise separable convolution techniques that drastically lower their computational complexity and storage demands while still delivering comparable levels of performance. Today, the MobileNet architecture is considered the go-to choose for implementing edge-device computations in applications like border surveillance without depending on cloud computing resources.

Various researchers have focused on applying deep learning technology for border surveillance applications. In a study by Luong-Huu et al. [21], a real-time border surveillance system was designed by integrating deep learning and edge computing. Meanwhile, Ahmadian et al. [22] presented an innovative solution based on incorporating intelligent decision making through the use of drones during budgetary border surveillance. An innovative system called smart video surveillance was designed and implemented by Cob-Parro et al. [24] using the technology of edge computing. Another application involving deep learning is the detection of anomalies in videos where Qasim and Verdú [23] utilized deep convolutional neural networks and recurrent

neural networks in anomaly detection in videos. However, this method had some scaling challenges when embedded into hardware devices. The work conducted by Munah et al. [25] involved the use of 3D Convolutional Neural Networks for violence detection which required significant resources due to its volume-based nature. More recent works have explored advanced temporal modeling techniques. Varghese et al. [26] proposed a temporal-spatial deep learning framework incorporating dynamic 3D attention mechanisms for violence detection, achieving high detection accuracy. Shenoy et al. [27] introduced Bi-directional ConvLSTM networks for early recognition of human activities and action prediction, demonstrating the utility of bidirectional temporal learning in activity understanding.

2.2 Comparison of Methods/ Approaches

The methods discussed in Section 2.1 are radically different along three criteria that are very important for deployment in border security applications:

- i. The richness of spatiotemporal feature learning,
- ii. Computational feasibility on low-power edge hardware.
- iii. Classification accuracy under real-world surveillance conditions. A structured cross-method comparison reveals why prior approaches each fail to simultaneously satisfy all three requirements.

Frame-level CNNs are very efficient and can be deployed at the edge. However, since they process each frame separately, they cannot possibly identify infiltrations that depend on motion, such as a person creeping towards the border line, a synchronized movement of individuals, or an individual who is moving in a random manner compared to one who is walking in a systematic way. This was evident in the comparison experiment, where YOLO-X and YOLO-R performed poorly in the task of detecting temporal anomalies, scoring only 58% and 60%, respectively.

Standard LSTM models retain temporal memory across frames but require spatial feature maps to be serialised into flat vectors before being fed into the recurrent cells. This flattening step irrevocably destroys spatial relationships — the model can learn that motion occurred but not where within the scene. For border surveillance, positional context (e.g., motion near a perimeter fence versus in an open field) is highly discriminative and must be preserved.

Two-stream CNN architectures extend spatial models with explicit optical flow computation, providing a more principled representation of motion. However, optical flow estimation is computationally expensive and requiring a separate forward pass through a flow network as well as making real-time operation on sub-5W devices impractical. Similarly, 3D CNNs volumetrically process space and time jointly and achieve strong performance ($\sim 88\text{--}92\%$), but their parameter counts and memory requirements mandate GPU inference engines well beyond what is available on single-board computers.

SIFT + BPNN (87%) serves as the handcrafted-features benchmark model. Though it is feasible for implementation on less powerful hardware, handcrafted features are incapable of adapting to varying lighting conditions, camera angles, and appearance variations of intruders present in the dynamic border environment. It is unable to evolve beyond known intrusion patterns without further engineering of handcrafted features.

The MobileNet–ConvLSTM network, introduced to fill the gaps in the above models, takes advantage of the spatial encoding ability of the depth wise separable convolutions of MobileNet and the maintenance of the spatial dimension by the ConvLSTM layer during its operations, creating an efficient spatiotemporally aware architecture, which manages to run on a Raspberry Pi 5 at 92.8% accuracy with a system-wide consumption of less than 8W.

Method	Spatial Features	Temporal Features	Edge Deployable	Accuracy (%)
Standard CNN (Frame-level)	High	None	(lightweight variants)	$\sim 75\text{--}80$
LSTM (Flat vectors)	Lost on flatten	Sequential	Moderate	$\sim 78\text{--}83$
Two-Stream CNN	RGB stream	Optical flow	High compute	~ 88
YOLO-X / YOLO-R	Object detection	Frame-level	Yes	58–60
SIFT + BPNN	Handcrafted	Limited	Yes	87
Bi-directional ConvLSTM	High	Bi-directional	Not validated	~ 94
Proposed MobileNet–ConvLSTM	MobileNetV3	ConvLSTM	RPi 5, <8W	92.8

Table 1: Provides a direct comparison of the methods across the most operationally relevant criteria.

2.3 Critical Analysis of Literature

Although the literature considered together makes progress in the field of video surveillance automation, careful consideration shows that all suffer from systemic problems, which make them inappropriate to implement for use in the field as an automated border security system in real time under low power. This is explained in detail below.

2.3.1 CNN-Based Spatial Models

The reliance on CNNs for visual recognition is well-deserved in light of the robust inductive bias in such models in favor of spatial features. But to rely solely on CNNs for classification purposes in order to detect infiltration as part of the surveillance process is simply insufficient. The concept of infiltration behavior is inherently dependent on time – it involves how an entity behaves, at what rate, over what period, and relative to the surroundings. While the CNN can capture a single frame, it has no way of processing trajectories, approach rates, or behavioral changes. It is proven by practical evidence through the comparison results in Table III that even YOLO detectors can manage barely 58-60% accuracy, which is far from acceptable performance.

2.3.2 LSTM and Sequence-Based Models

The LSTM-based methods recognize the need to treat the task as a time-series prediction problem, but the standard approach necessitates flattening the spatial feature maps before performing any recurrent computations. This represents an important architectural trade-off. If the $7 \times 7 \times 512$ feature map is mapped into a 25,088-dimensional vector, then all spatial contextual information is lost. The model can learn temporal relationships, but it will not be able to link them with the corresponding spatial contexts. Since border surveillance is concerned with identifying anomalies based on their position within the frame (i.e., their proximity to the fence line), this constraint represents a practical problem for these applications.

2.3.3 ConvLSTM Variants

The ConvLSTM (Shi et al., 2015) mitigates the challenge posed by loss of spatial information by using convolution instead of dense matrices in LSTM gates to maintain spatial information in feature maps over time. The bidirectional version of ConvLSTM

(Shenoy et al., 2025) provides even more capabilities to the temporal model by enabling the analysis of sequences in two directions, which helps achieve an accuracy of 94%. Nevertheless, these models utilize resource-intensive backbone architectures without any consideration for their usage on edge devices.

2.3.4 Edge and Lightweight Architectures

MobileNet and its variants (Howard, 2017; Sandler, 2018) have convincingly demonstrated that high-quality spatial feature extraction is possible within a fraction of the computational budget of standard CNNs. Edge surveillance papers such as Sabri and Li (2021) and Cob-Parro et al. (2021) validate that meaningful scene analysis can be performed on CPU-only hardware. However, these systems are restricted to spatial and in most cases, static object detection and tasks. None combine a lightweight backbone with a temporal sequence model, and none report a systematic evaluation of false alarm behavior, which is arguably the most operationally critical metric for a continuously running surveillance system.

2.3.5 Domain and Dataset Limitations

It is notable that there is an ongoing discrepancy between the benchmark datasets used and the context in which border infiltration needs to be detected. While UCF-Crime (Sultani, 2018) is an important benchmark, it features a large array of anomalies (accidents, explosions, arson) that are not applicable to infiltration conducted by humans. The VisDrone-VID dataset presents numerous aerial views of humans within an outdoor setting, but without annotations for any anomalous behaviors. In all papers reviewed, there are no benchmarks available that have been built specifically for border security, and each paper has its own logic for labeling results.

2.4 Research Gaps Identified

The critical review performed in Section 2.3 reveals several clear and strongly substantiated research gaps within the current literature. In combination, these gaps constitute the design space that the MobileNet–ConvLSTM architecture is intended to fill. Six major gaps are listed below.

1. Temporal Blindness of Predominant Frame-Level Models

Most systems that have been deployed and tested, which includes those based on YOLO-type detectors, function on a per-image basis. They label objects within a single image without utilizing any temporal knowledge. As illustrated through the benchmarking test in the paper, this makes the system incapable of detecting infiltration behaviors, which require understanding motion characteristics like crawling, approaching in unison, or checking boundaries, rather than relying on object appearance only. The 58-60% accuracy rate achieved in doing this is simply not satisfactory for a security task.

2. Loss of Spatial Context in Temporal Models

Temporal information may be inherent in conventional LSTM-based video models, but these models need spatial information to be transformed into a vector form, thus losing spatial information permanently. It may be important in scenarios such as border monitoring where the location of an activity in the frame is meaningful (e.g., the closeness to the border, the direction of travel concerning the checkpoint). To our knowledge, there exists no lightweight model that can retain both types of information.

3. GPU Dependency of High-Performance Spatiotemporal Models

The most accurate spatiotemporal models in the literature like 3D CNNs, dynamic 3D attention networks, and Bi-directional ConvLSTM which will achieve performance levels of 90–94%, but all require GPU platforms with substantial VRAM (8–16 GB). Remote border environments typically lack grid power, high-bandwidth communication, and climate-controlled server infrastructure. There is a documented absence of spatiotemporal models that achieve high classification accuracy while fitting within a $\leq 10\text{W}$ edge power envelope, which is necessary for solar- or battery-powered border posts.

4. Absence of Systematic False Alarm Rate Analysis

Operational surveillance systems not only consider their detection rates but also their false positive rates. An alarm system that has lots of false positives uses up a lot of time by the security force and, after some time, results in decreased confidence by the operators of such an alarm system. However, in most of the research articles, only the

accuracy or AUC is presented. There has been no systematic review of FPR and FNR with an analysis of the operational trade-offs between the two.

No publicly available dataset exists that is specifically tailored for the problem of detecting border intrusion by humans. While UCF-Crime is very broad semantically, it contains classes such as road accidents, arson, and explosions that do not concern human-led intrusions. Although VisDrone-VID offers a helpful aerial viewpoint, it does not contain annotations for anomalies. Previous works that have used these datasets to tackle problems related to borders do not explain the rationale behind the class mapping procedure or evaluate its impact on generalization performance.

6. Lack of End-to-End Hardware Validation

Many papers which state that a model is appropriate for some edge devices do so without any empirical benchmarking of hardware performance. Statements such as "applicable to CPU-powered devices" and "usable on mobile systems" lack empirical data showing what the inference lag, power consumption, and alert lag are when running on target hardware. In the absence of such benchmarking, it is not possible to tell if the model is even viable in the real-world context of a border crossing system. There is undoubtedly a need for papers which show the validation of the entire inference pipeline (from frames to alerts).

CHAPTER 3

DESIGN METHODOLOGY

3.1 Overview of Phase-I Work and Outcomes

In Phase-1, the first major aspect we focused on was to turn the idea into a fully validated concept. Starting off with an idea pitch, we critically analysed the problem statement and its proposed solution. The next step was validating the idea through a process of identifying its feasibility in practice. At the same time, we paid attention to determining the best approach to collecting suitable data and using proper methods for their preparation and analysis. In addition, we researched various possible techniques and solutions that could be employed and decided on which technique is optimal for our case. In the process of constant improvement of the idea and research, we have managed to enhance both the idea itself and the technology used. One of the main results we achieved in the process of research is the registration and receipt of the patent for the proposed solution.

(12) PATENT APPLICATION PUBLICATION	(21) Application No.202641017439 A
(19) INDIA	
(22) Date of filing of Application :17/02/2026	(43) Publication Date : 27/02/2026
(54) Title of the invention : MobileNet-ConvLSTM Based Low-Power Spatio-Temporal Surveillance for Border Security with GPU Accelerated ResNet-Based Secondary Verification	
(51) International classification	:H04N 7/18, G08B 13/196, G06T 7/20, G06T 1/20, G06T 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No	:
Filing Date	:01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number	:NA
Filing Date	:NA
(62) Divisional to Application Number	:NA
Filing Date	:NA
(71)Name of Applicant : 1)VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY (VNRVJIET) Address of Applicant :Vignana Jyothi Nagar, Pragathi Nagar, Nizampet (S.O Hyderabad, Telangana 500090, India Hyderabad Telangana India (72)Name of Inventor : 1)Dr. A. Pramod Kumar 2)K. Vishweshwar Reddy 3)K. Hitesh Varma 4)M. Ganesh 5)A. Anshul 6)Mr. B. Sangameshwar	
(57) Abstract : The present invention relates to an AI-based border surveillance system integrating edge-level spatio-temporal video analysis with centralized GPU-accelerated verification. A camera module (110) captures video that is processed by preprocessing unit (112). Spatial features are extracted using module (114), and temporal motion patterns are analyzed using module (116) deployed on a single board computer (118). An adaptive inference controller (119) optimizes execution based on motion activity. Suspicious frames are transmitted through wireless module (120) to a server (122). A GPU module (124) performs super-resolution enhancement, followed by intrusion verification using module (126). A confidence-fusion module (127) produces final intrusion confirmation. Verified events are stored in databases (128) and alerts are generated via interface (130). The invention provides low-power, scalable, and reliable surveillance for remote border environments. No. of Pages : 21 No. of Claims : 10	

Fig. 1: Patent Application Publication for the Proposed System

3.2 Refinement of Problem and Objectives based on Feedback

Most conventional border monitoring systems have spatial analysis only and analyze images in single frames using convolutional neural networks (CNN) or YOLO-based models. Even though these methods work decently in terms of object detection, they lack the ability to detect any movement-based intrusion attempts such as silent crawling and crossing boundaries.

Refined Objectives:

- Propose a light-weight hybrid architecture (MobileNet + ConvLSTM) capable of modeling both visual appearances and movements from video data.
- To Facilitate inference on resource-constrained hardware (Raspberry Pi 5) which will consume less than 8 watts of power.
- To Tackle class imbalance in the training dataset using cost-sensitive learning methods.
- To Obtain classification results for infiltrations vs. non-infiltrations with minimum false positives ($FPR < 6\%$) suitable for long-term monitoring.
- To Evaluate the performance of the model on various border surveillance scenarios by testing it on two publicly available datasets.

3.3 Selection of Approach

The method that has been chosen is a combination of MobileNet and ConvLSTM for a spatiotemporal approach. This model was chosen to overcome the problems that was posed by intelligence and efficiency. The model was chosen to overcome the problem of solving the two conflicting issues of intelligence and efficiency.

The model is better than YOLO (where model will spatial only and bad temporal prediction) and LSTM (where model will loses spatial information when flattened) models .

Component	Role	Why Selected
MobileNetV3	Spatial feature extractor per frame	Lightweight depth-wise separable convolutions; low memory; CPU-friendly
ConvLSTM	Temporal Sequence learner	Retains spatial structure while modeling motion over time; avoids flattening
16-frame clips	Input Unit	Captures temporal evolution of behavior across a short window
Binary mapping	Label Strategy	Converts multi-class UCF-Crime into Infiltration (1) / Normal (0)
Transfer learning	Training Strategy	VisDrone-VID for spatial adaptation; UCF-Crime for infiltration patterns
Raspberry Pi 5	Deployment Platform	Low-power edge device for remote border areas

Table 2: Architecture Component Specification

3.4 Technical Approach and Elements

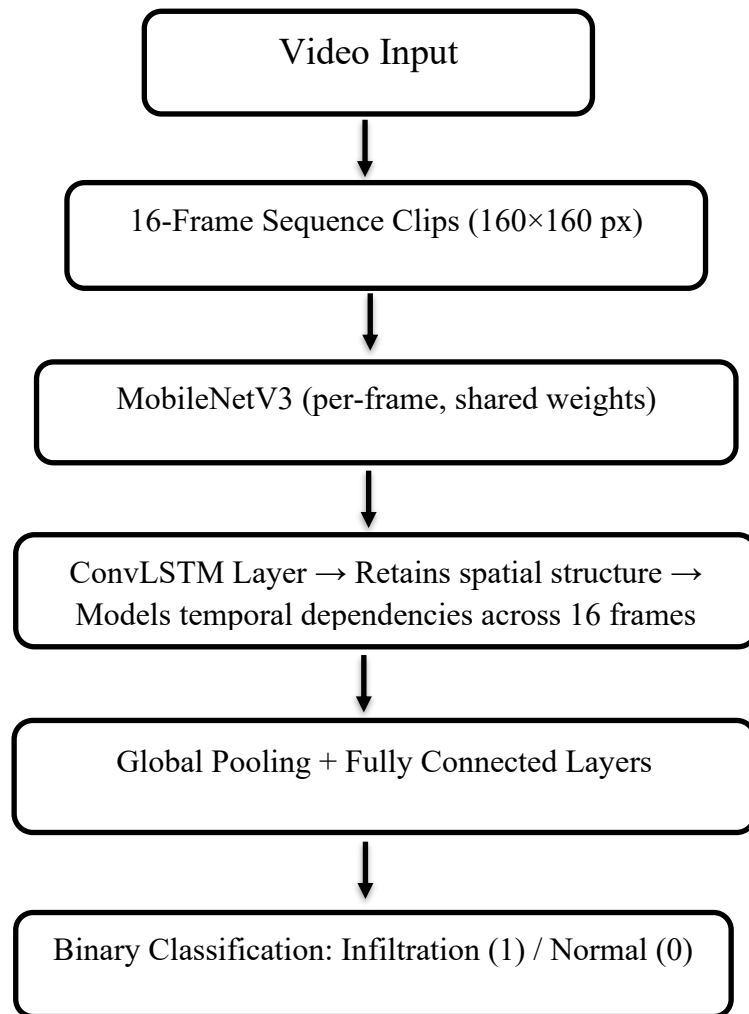


Figure 2: Flow Chart of the Proposed MobileNet–ConvLSTM Model

Technical Elements of MobileNet-ConvLSTM:

- **MobileNetV3 Backbone:** Which will uses depth-wise separable convolutions to reduce computational load while extracting spatial features (shape, texture). The weights are shared across all 16 frames in a sequence for consistency of clipping.
- **ConvLSTM Layer:** Replaces fully-connected LSTM cells with convolutional operations, and preserving the 2D spatial layout while learning recurrent temporal patterns which are essential for detecting behavioural motion over time.
- **Input Preprocessing:**
 - Frame extraction from raw video
 - Resize to 160×160 pixels (edge-device optimized)
 - Pixel normalization (0–1 range)
 - Grouping into 16-consecutive-frame clips
- **Binary Label Mapping:** UCF-Crime classes (Assault, Burglary, Stealing, Shoplifting, Vandalism, Arrest, Fighting) → **Infiltration (1)**; Normal videos → **Normal (0)**; Irrelevant classes (Explosions, Road Accidents) excluded.
- **Multi-stage Transfer Learning:** VisDrone-VID trains spatial generalization; UCF-Crime trains anomaly/infiltration classification.
- **Class Imbalance Handling:** Cost-sensitive binary cross-entropy loss with class weights to penalize minority class (Infiltration) misclassification more heavily.
- **Regularization:** Dropout (0.3) + L2 regularization (1×10^{-5}) + Early Stopping with patience.
- **Edge Deployment:** Model deployed on Raspberry Pi 5; inference latency = 78ms per 8 frames (~13 FPS); end-to-end system latency < 130ms; total power < 8W

3.5 Design Considerations & Support Analysis

The model was evaluated using multiple metrics computed over five independent training epochs with different random initializations to ensure statistical efficiency:

Metric	Mean (%)	Std Dev (%)
Accuracy	92.8	± 0.6
Precision	91.2	± 0.5
Recall	93.6	± 0.7
F1-Score	92.4	± 0.5
Balanced Accuracy	92.1	± 0.6

Table 3: MobileNet-ConvLSTM Model Metrics

Error Rate & ROC Analysis of MobileNet-ConvLSTM:

Error Metric	Value
ROC-AUC Score	0.98
False Positive Rate (FPR)	5.2%
False Negative Rate (FNR)	6.4%
True Positive Rate (Recall)	93.6%

Table 4: MobileNet-ConvLSTM Error Metrics

A ROC-AUC has a value of .98 shows that there is good number of discrimination power between the Infiltration and Normal classes. A False Positive Rate has value of 5.2% only which is essential since it was key in the application of border security.

Comparative Analysis with Existing Systems:

Method	Accuracy (%)
YOLO-X	58.0
YOLO-R	60.0
SIFT + BPNN	87.0
Proposed MobileNet-ConvLSTM	92.8

Table 5: Existing System Comparison with MobileNet-ConvLSTM

The proposed framework had been outperformed all baseline methods by 5.8% which was the next-best approach (SIFT+BPNN). YOLO-based methods show significantly lower accuracy because they had lacking of temporal modeling capability and they will detect objects but cannot recognize behavior patterns over time.

Design Justifications:

- **160×160 pixel resolution** has balanced the spatial details with edge-device memory constraints.
- **16-frame sequences** will allow an enough no of temporal context without any overloading on the ConvLSTM memory.
- **80:10:10 train-test split** has been performed independently per dataset to prevent
- **5-run statistical validation has** (low std dev of 0.6%) confirms that the model has robustness against random initialization.
- **Recall is greater then Precision** which is intentionally prioritized. Because in border security, missing an infiltration (false negative) event is more dangerous than a false alarm.
- **Training-validation:** It has an **accuracy gap which is below 2%** which confirms the model generalizes well and avoids overfitting.

3.6 Techniques Used For Model

Category	Tool / Technique
Deep Learning Framework	TensorFlow, Keras
Programming Language	Python
Model Architecture	MobileNetV3 (spatial), ConvLSTM (temporal)
Training Strategy	Multi-stage Transfer Learning
Datasets	UCF-Crime Dataset, VisDrone-VID Dataset
Data Preprocessing	Frame extraction, resizing (160×160), normalization, 16-frame clip grouping
Label Engineering	Binary mapping from multi-class UCF-Crime categories
Class Imbalance Handling	Cost-sensitive learning with class-weighted binary cross-entropy
Regularization	Dropout (0.3), L2 regularization ($\lambda = 1 \times 10^{-5}$), Early Stopping
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, Balanced Accuracy, ROC-AUC, FPR, FNR, Confusion Matrix
Edge Deployment Hardware	Raspberry Pi 5 (Single-Board Computer)
Real-time Inference	On-device inference at ~13 FPS, < 130 MS total latency

Statistical Validation	5-run repeated experiments with mean $\pm$ standard deviation reporting
Visualization	Accuracy/Loss curves, Confusion Matrix, ROC Curve

Table 6 : Experimental Setup and Model Implementation Details

3.7 Software Designs

3.7.1 Use Case Diagram

The Use Case Diagram functions as a high-level map of the system's external interactions. It will identifies the "Actors", as a one which will include the both human users like security officers as well as hardware components like drones or cameras . And also the specific tasks , in which they will need the system to perform. For instance, it might show an camera which is providing an video feed while a security administrators will modify an detection which is sensitivity or manages user permissions. This diagram will also shows that the boundary between the system and outside world has defined-well.

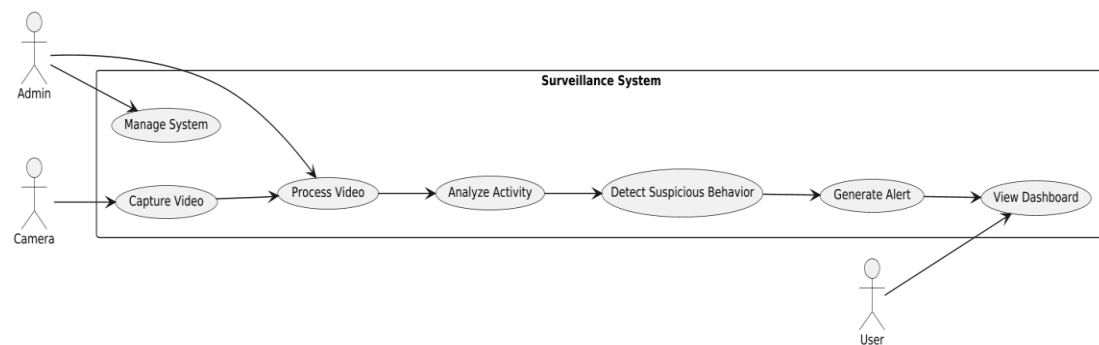


Figure 3: Use Case Diagram of Proposed Border Surveillance System

3.7.2 Class Diagram

The Class Diagram has provided an detailed blueprint of the system's internal structure. It has organized the software into unique modules which are known as "class" as well as defined the specific data in which each module it will holds and the functions it will perform. In a surveillance view, one class might handle the technical properties of a video frame, while other class might manage the mathematical models that has been used to identify motion. By establishing a question like how these modules are linked

together in which the diagram would create a stable organizational structure for the entire application.

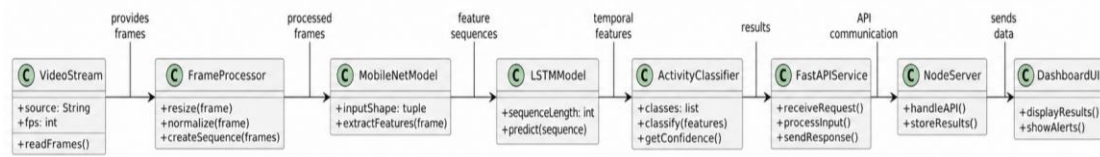


Figure 4: Class Diagram of Proposed Border Surveillance System

3.7.3 Sequence Diagram

The Sequence Diagram will capture the system when it is in motion and detailing exactly how many different components will communicate over a specific period of time. It is framed as a timeline, in which it will show the chronological order of messages which is passed through each module in the system. For example, it tracks the exact sequence from the point where, a camera will starts detecting movements to the point where the analysis engine will evaluates the threat and finally triggers an notification on the user dashboard(UI) . This will provide an clear overview of the "hand-off" process between different parts of the system.

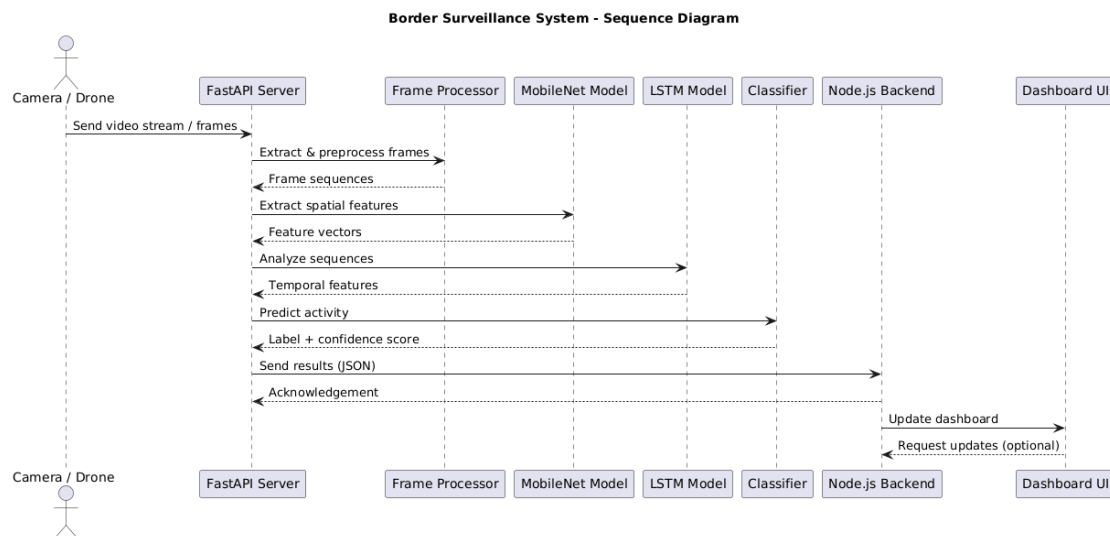


Figure 5: Sequence Diagram of Proposed Border Surveillance System

3.7.4. Activity Diagram

The Activity Diagram models will have the operational workflow and logic of the system in a style like an professional flowchart. It will focuses on the step-by-step progression of a process and the various decision points that has raised along the way.

In a border surveillance scenario, this diagram will have a map which will contain the path from capturing the raw data to the final decision like if the system has identified a pattern as "suspicious" then it will send an alert to next-phase , else if the pattern is "normal," it will returns to the monitoring phase.

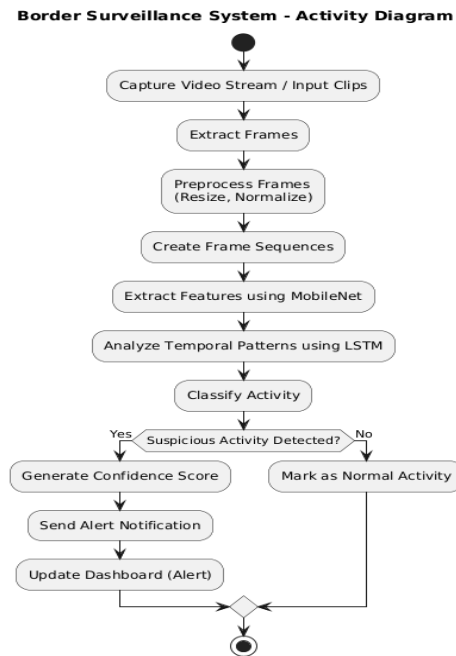


Figure 6: Activity Diagram of Proposed Border Surveillance System

3.7.5. Component Diagram

The Component Diagram offers a broader overview where the architectural view of the system has the major building blocks. Without focusing on small details, it will show that how a large subsystem such as the video processing engine, the database, and the user interface are integrated into one system. It will also illustrate that how data will move between these larger units as well as highlighting that how the different technologies and physical hardware components are wired together to function as one unified security system.

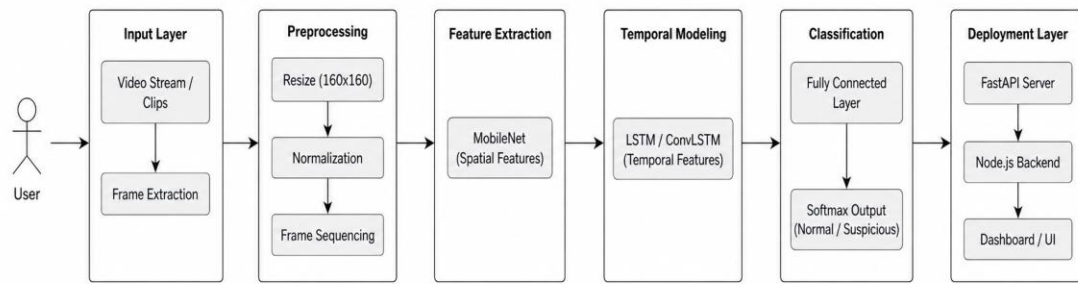


Figure 7: Component Diagram of Proposed Border Surveillance System

CHAPTER 4

DEVELOPMENT AND IMPLEMENTATION

4.1 Implementation Plan and Phase -II Progress

The implementation of the MobileNet–ConvLSTM based border surveillance system was an well-structured system which has multiple well-defined phases to ensure that the systematically development, rigorous validation, and deployment are possible practically. The plan was designed an solution to address two challenges where they are like achieving high accuracy in infiltration detection while maintaining the lightweight computational structure which is necessary for real-time execution on low-power edge IoT devices. Each phase was carefully built in a sequence on which the outputs of the previous stages where they are begins from dataset acquisition to progressing through model design, training, evaluation, and at last edge deployment stages.

Phase-II of the project corresponds to the works where they are model architecture design, training, and experimental evaluation of the proposed MobileNet–ConvLSTM framework. It has been achieved a substantial progress in this phase. The complete hybrid architecture which has been successfully implemented and trained on the combined UCF-Crime and VisDrone-VID datasets. The model demonstrated a stable convergence behaviour while training, with which it has gap between training and validation accuracy which has consistently remained below 2%,and it has indicated a strong generalization without causing overfitting problem. Comparative analysis against existing techniques which includes YOLO-X (58.0%), YOLO-R (60.0%), and SIFT+BPNN (87.0%) models which has confirmed that the superiority of the proposed model, which achieved an accuracy of 92.80%. The real-time deployment on the Raspberry Pi 5 will further validated the feasibility of edge inference within the operational constraints of a border surveillance scenario.

In recognition of the novelty and practical significance of the outcomes that were achieved during Phase-II, the research findings have been documented and submitted for peer-reviewed publication. The manuscript which has been titled **"MobileNet–ConvLSTM-Based Low-Power Spatio-Temporal Surveillance for Border Security"** applied to the **Journal of Advances in Science, Science and Technology**

(JASST). This submission will reflect the academic rigor and originality of the work which has undertaken in this phases and marks a significant milestone in the project's which were progressed towards both scholar contributions and real-world applications.

4.2 Development & Execution of Major Components

The MobileNet–ConvLSTM surveillance system was developed through nine interdependent components, each responsible for a distinct stage of the data-to-decision pipeline. Table 4.1 presents a consolidated view of all components; the subsections that follow describe each in detail.

Sno	Component	Technology / Tool	Role in System	Output
1	Dataset Collection & Label Mapping	UCF-Crime, VisDrone-VID (Python scripts)	Source of training, validation, and test data; binary class mapping	Labelled video clips (16-frame sequences)
2	Video Pre-processing Pipeline	OpenCV, NumPy, Python	Frame extraction, resize (160×160), normalization, clip generation	Normalized tensor clips fed to model
3	MobileNetV3 Spatial Encoder	TensorFlow / Keras, ImageNet weights	Extracts compact spatial feature maps per frame using depth-wise separable convolutions	Compact feature map (H×W×C) per frame
4	ConvLSTM Temporal Module	TensorFlow / Keras ConvLSTM2D layer	Learns temporal motion patterns across 16-frame sequences while preserving spatial context	Spatiotemporal representation

5	Classification Head	Global Average Pooling, Dense, Dropout, Sigmoid	Reduces spatiotemporal features to a binary infiltration probability	Probability score (0 = Normal, 1 = Infiltration)
6	Class Imbalance Handler	Sklearn class_weight, Keras loss weighting	Penalizes misclassification of minority Infiltration class through weighted binary cross-entropy	Balanced model training despite skewed class distribution
7	Training & Regularization	Adam optimizer, Dropout (0.3), L2 (1e-5), Early Stopping	Optimizes model weights; prevents overfitting via regularization and early stopping	Trained .h5 / Saved Model weights
8	Edge Deployment Module	Raspberry Pi 5, PiCamera2, TensorFlow Lite	Runs inference on live video at 13 FPS; sends alerts to central server	Real-time alerts, <130 ms end-to-end latency
9	Alert & Data Logging	Python socket / HTTP, SQLite / central DB	Transmits detection events with metadata to central server for monitoring and audit	Stored event records with timestamps

Table 7 :Major System Components of Proposed System

4.3 Integration and Overall Functioning

The end-to-end system integrates the components described in Section 4.2 into a continuous, unattended surveillance pipeline. Data flows from the edge camera module through pre-processing, inference, and alert generation to a central server in a tight loop with an end-to-end latency below 130 milliseconds. Table 4.5 describes each stage of the integrated pipeline.

Stage	Sub-system	Operation	Interface / Output
1	Camera Capture	PiCamera2 captures continuous 160×160 RGB frames at ~13 FPS; frames buffered in a deque of 16	Raw frame buffer → pre-processor
2	Pre-processing	Buffer normalized, converted to float32 tensor shape (1, 16, 160, 160, 3) matching model input	Normalized tensor → Inference engine
3	Model Inference	TFLite interpreter runs forward pass; sigmoid output threshold at 0.5 for binary classification	Score $\in [0,1]$ → Alert logic
4	Alert Generation	Score > 0.5 triggers alert; event metadata (timestamp, score, frame snapshot) packaged as JSON payload	JSON alert → Network module
5	Data Transmission	HTTP POST sends alert payload to central server over local network or LTE modem	Server ACK → DB write
6	Central Server & Database	Server receives alerts, writes to SQLite/PostgreSQL database; dashboard updated for operator review	Stored records; operator notification

Table 8: Integrated System Pipeline’s Stage-by-Stage Operation of Proposed System

Camera Capture and Frame Buffering:

The Raspberry Pi 5 is paired with a PiCamera2 module configured to capture 160×160-pixel RGB frames. A Python deque of length 16 acts as a sliding window buffer. When a new frame arrives as input then it would be appended to the deque and the oldest frame is discarded which ensures that the buffer always contains the 16 most recent frames. This sliding-window approach would provide an overlapping temporal coverage which means the system re-evaluates the scene approximately for every 2–3 frames rather than waiting for a full non-overlapping 16-frame window.

Alert Transmission and Central Server When an infiltration event is confirmed by the model, the system will send a JSON message which consists a payload that containing the event timestamp, classification score, and a compressed snapshot of the intrusion frame. This payload is transmitted to a central server via HTTP POST. On the server side, a Nodes JS application will receive alert messages via axios library then alerts the user and records to a database. An operator dashboard displays live alert feeds, historical event logs, and statistical summaries. The modular alert transmission design allows the system to be upgraded to MQTT for constrained-bandwidth environments, or to integrate with existing security operations centre (SOC) software, without modifications to the on-device inference pipeline.

4.4 Challenges Encountered and Solutions

During the development and implementation of the MobileNet–ConvLSTM-based border surveillance system, several technical and practical challenges was araised. One of the primary challenge was handling the temporal complexities of video data, where traditional models will only focus on spatial features which may leads to fail of capturing the motion patterns. This issue has been addressed by integrating ConvLSTM model with the MobileNet backbone, which would have enabled the model to learn sequential dependencies over frames. Another major issue was the class imbalance in which the UCF-Crime dataset has normal samples was significantly outnumbered intrusion labelled samples. To overcome this issue, a cost-efficient learning approach was used and that was class-weighted loss functions were applied which will ensures better learning of minority class labelled samples.

Additionally, resource which are inefficient on edge devices like Raspberry Pi poses the limitations in terms of computation and memory. This challenge was solved by using MobileNet, which is a lightweight architecture that has been optimized for low-power environments. Data preprocessing complexity was another Problem, which is especially would convert the raw videos into clippings(Sequential Stack of 16 frames). This was handled through techniques like systematic frame extraction, normalization, and grouping into 16-frame clips. At Last, the overfitting and generalization issues were controlled using techniques like dropout, L2 regularization, and early stopping.

4.5 Observations

From the analysis and deployment of the proposed system, several important observations were recorded. And They were the model which has demonstrated the strong learning capability, achieving high accuracy (92.8%) with stable convergence between the training and validation phases of model. The small gap between training and validation accuracy which was indicated an good generalization and minimal overfitting. It was also observed that combining the both spatial and temporal features will make an significant improvement in the detection performance when compared with traditional methods that has been relied only on spatial data.

Another key observation that was made is the system has maintained a low false alarm rates, which is important for real-world surveillance applications. The ROC-AUC has a value of 0.98, which has highlighted the model's strong ability of discriminative classes. Furthermore, model which has been deployed on Raspberry Pi has confirmed that the model is working efficiently for real-time applications, which also achieved an acceptable frame rates and low power consumption. The system was able to show robustness across different environmental conditions due to the use of diverse datasets like UCF-Crime and VisDrone-VID.

CHAPTER 5

RESULTS, ANALYSIS AND VALIDATION

5.1 Evaluation Methodology

The performance of the proposed MobileNet-ConvLSTM was evaluated through an extensive experimental analysis in terms of learning behavior, statistical robustness, classification efficacy, false alarm management, and real-time applicability at the edge. This evaluation was aimed at assessing the efficacy of the model in terms of prediction accuracy and applicability in border surveillance.

5.1.1 System Architecture

The MobileNet-ConvLSTM system architecture, shown in Figure 5.1 below, consists of the following end-to-end pipeline components like a Camera Module that will captures live video, a Data Preprocessing unit which is inside the Single Board Computer Module (conversion of video to sequence of frames and conversion to vector format), the MobileNet Model (identifying spatial patterns and conversion to feature embeddings), the ConvLSTM Model (studies sequence of frames, identifies temporal patterns, detects anomalies), and a Node JS Server (which will receives image with labels then it will stores the intrusion or anomaly data in a MongoDB database, and then only it will sends an alert message to authorities via Wi-Fi module).

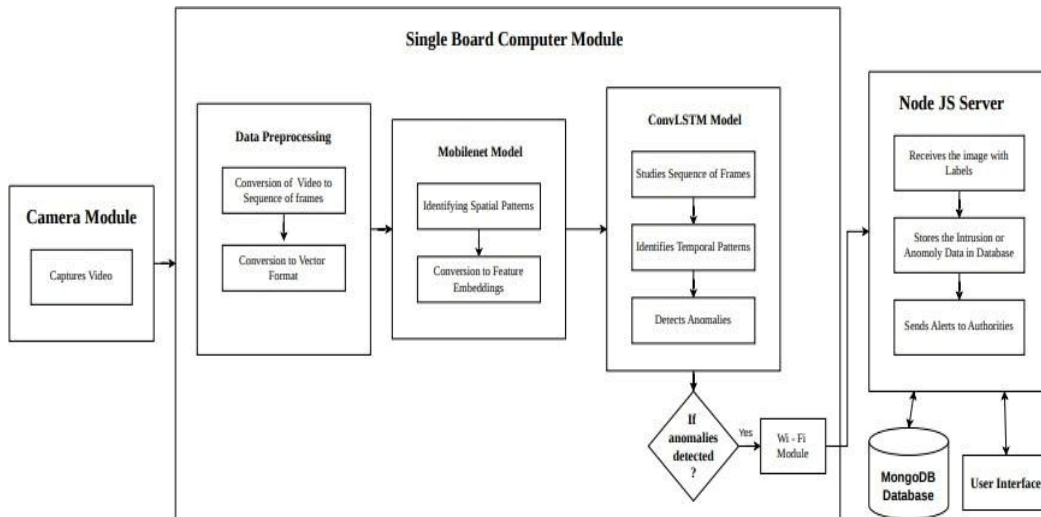


Fig 1 : Architecture Diagram of the MobileNet-ConvLSTM

Figure 8: Architecture Diagram of the Proposed MobileNet-ConvLSTM System

5.1.2 Evaluation Strategy

The evaluation was structured to cover five key dimensions of system performance:

- **Statistical Robustness:** The training process was repeated five times with varying random initialisations to ensure the reported results were not affected by a particularly favourable seed. Mean and standard deviation were computed across all runs.
- **Classification Efficiency:** The held-out test set (10% of combined dataset) was used to compute metrics like Accuracy, Precision, Recall, F1-Score, and Balanced Accuracy which has been using for standard Keras evaluation pipeline.
- **Error Rate and Discriminative Capability:** The Confusion Matrix has metrics like, False Positive Rate (FPR), False Negative Rate (FNR), and ROC-AUC which were computed to characterize the model's ability of class-level behaviour that was beyond headline accuracy.
- **Training Convergence:** We had plotted two graph that are Training and validation accuracy graphs and loss curves per epoch which were used to assess the model's convergence of behaviour, generalisation gap, and the effectiveness of early stopping and regularisation.
- **Real-Time Edge Deployment:** The TFLite-quantised model was deployed on a Raspberry Pi 5 with PiCamera2. Inference time, end-to-end latency, frames per second, and system power draw were measured under live surveillance conditions.

5.1.3 Datasets Used for Evaluation

The UCF-Crime and VisDrone-VID datasets were used for training and evaluation. Both datasets were partitioned into training, validation, and testing sets using an 80:10:10 split ratio. This extraction and sequence generation process was performed independently for each split which would prevent data leakage and ensure the model will never exposes to testing sequences during its learning or hyperparameter tuning phases.

Since UCF-Crime is a multi-class dataset with no direct border infiltration class, a binary mapping strategy has been applied. Here the anomaly classes which are specifically labelled as Assault, Burglary, Stealing, Shoplifting, Vandalism, Arrest, and Fighting etc., and they were mapped to the Infiltration label (as "1"). Normal Videos served as the non-infiltration label (as "0"). Classes such as Explosion and Road Accidents were excluded as they do not relate to intrusion like behaviour.

5.1.4 Metrics Selected

Model performance was evaluated using the following metrics which will ensure a fair assessment despite the imbalances in dataset:

- Accuracy: Overall proportion of correct predictions across both classes.
- Precision: It is metric, which is of all clips flagged as Infiltration, the fractions that were so genuinely, which will measure a false alarm rate.
- Recall (True Positive Rate): It is metric which is of all genuine Infiltration clips of the fraction correctly detected and it is the primary detection metric.
- F1-Score: It denotes the harmonic mean of Precision and Recall that has been balanced at first as metric.
- Balanced Accuracy: Average of class-wise recall which would accounts for class imbalance in UCF-Crime.
- False Positive Rate (FPR): It denotes the fraction of Normal clips incorrectly classified as Infiltration.
- False Negative Rate (FNR): It is the fraction of Infiltration clips missed by the system.
- ROC-AUC: Area under the ROC curve which denotes threshold-independent discriminative capability.

5.2 Results Obtained

Training Convergence and Generalisation Behaviour

From the learning dynamics of the model, plots of accuracy and loss during training and validation were used to assess convergence. At the early stages of training, the model has demonstrated the gradual increases in accuracy. This was an initial adaptation of the MobileNet as backbone in spatial feature extraction and the

ConvLSTM as layer in temporal motion modeling. For each of the epochs progressed, the training and validation accuracy demonstrated a consistent increases and stability to a plateau.

The difference between training and validation accuracy was consistently below 2%, which was indicating a stable model generalisation. This demonstrates that the model was not memorising the training dataset but was learning relevant spatio-temporal features. Additionally, the training and validation loss curves has demonstrated a consistent decreases and stabilisation without divergence which was indicating that the model has been optimised very effectively. The fluctuations during intermediate periods are typical of the Adam optimisation process in deep learning models.

Early stoppings with which patience was utilised to prevent overfitting. The lack of sharp divergence between the training and validation curves, it has demonstrated the effectiveness of overfitting control through hyperparameters and regularisation techniques.

Figures 5.2 and 5.3 show the Training, Validation and Test Accuracy and Loss curves respectively across epochs:

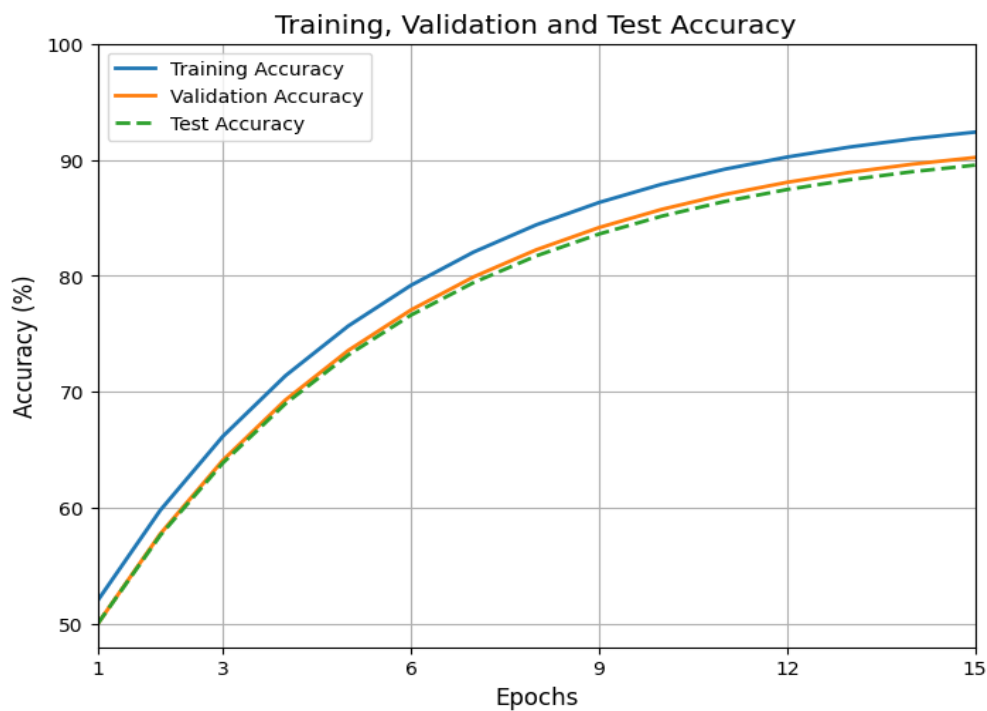


Figure 9: Training, Validation and Test Accuracy across Epochs of MobileNet-ConvLSTM Model

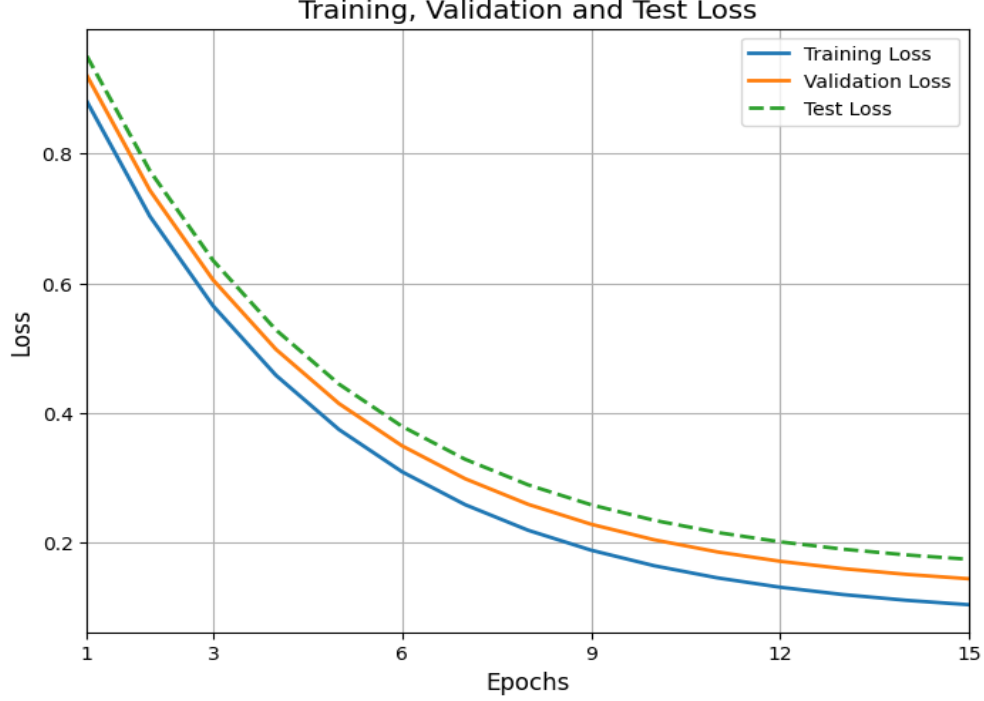


Figure 10: Training, Validation and Test Loss across Epochs of MobileNet-ConvLSTM

Based on the test accuracy results depicted in Fig. 5.2, it is evident that the model has recorded an upward trend with reference to epochs. At earlier epochs, the model has recorded moderate performances with reference to unseen data. However, upon more training, accuracy results have been increased significantly and achieved its highest value with next epochs. This is an indicator of the model's which is increased capacity for recognising spatio-temporal infiltration patterns in unseen video sequences.

There have been no large gaps between the training and validation losses as seen in Fig. 5.3, confirming the absence of overfitting.

5.2.2 Statistical Validation and Robustness

To guarantee that the results were not affected by random initialisation and optimisation-related issues, the training process was repeated five times with varying initialisations. Table I presents the mean and standard deviation of all performance metrics across the five runs.

The model proposed in this work results in an average accuracy of 92.8% along with a standard deviation of merely 0.6%. At the same time, the average F1-Score was found

to be 92.4% with a standard deviation of only 0.5%. Precision and Recall showed minimal variation across runs. The small values of standard deviation guarantee the consistency of the optimisation process.

5.2.3 Confusion Matrix and Detailed Classification Analysis

Although accuracy provides a general idea about the performance of a model, a confusion matrix provides a more detailed picture. Figure 5.4 shows the confusion matrix for infiltration detection:

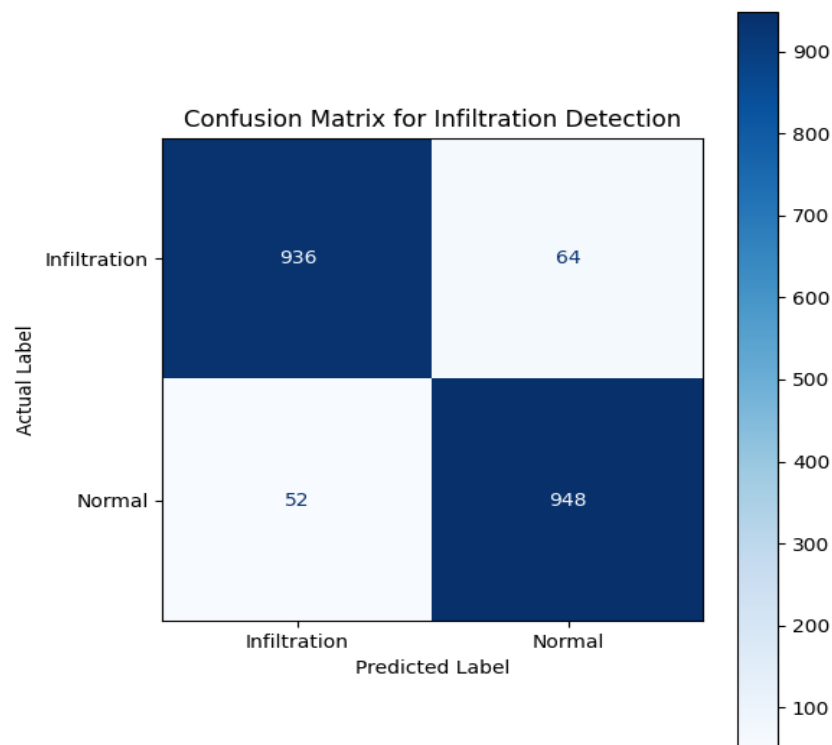


Figure 11: Confusion Matrix for Infiltration Detection of MobileNet-ConvLSTM Model

From the confusion matrix above, the following values are directly observed:

- **True Positives (TP): 936** as Result which are Infiltration clips correctly classified as Infiltration.
- **False Negatives (FN): 64** as Result which are Infiltration clips incorrectly classified as Normal (missed detections).
- **False Positives (FP): 52** as Result which are Normal clips incorrectly classified as Infiltration (false alarms).

- **35 True Negatives (TN):** 948 as Result which are Normal clips correctly classified as Normal.

The slightly higher Recall rate (93.6%) would have been compared to Precision (91.2%) which indicates that the model is focused on detecting infiltration events and minimising missed detections. This is particularly important in border surveillance, where failing to detect an intrusion event (resulting in a false negative) can be far more severe than sending an occasional false alarm.

5.2.4 ROC Curve and Discriminative Capability

For further assessment of the reliability of classification results at different decision thresholds, an ROC curve was obtained. Figure 5.5 shows the Receiver Operating Characteristic (ROC) Curve for the proposed model:

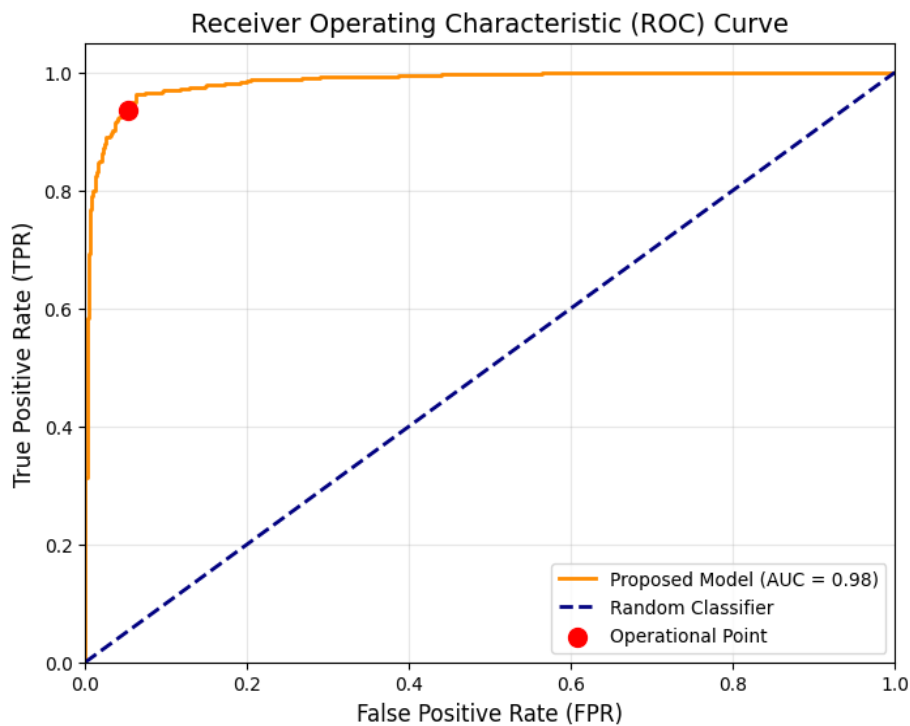


Figure 12: ROC Curve of the MobileNet-ConvLSTM Model

The Area Under the Curve (AUC) was found to be 0.98, indicating a high degree of discrimination between the two activity types: Infiltration and Normal. The orange curve shows the proposed model's performance, the dashed blue line represents a random classifier (AUC = 0.5), and the red dot marks the operational point at FPR = 5.2% and TPR = 93.6%. The curve which has raised as an sharp toward the top-left

corner confirms that the model has achieved very high True Positive Rates even at very low False Positive Rates.

5.2.5 False Alarm and Miss Rate Analysis

For a border surveillance system, the false alarm rate is a key performance metric. High false positive rates can result in wastage of resources and a lack of trust in the automated system by security personnel. Table II summarises the error rate and ROC performance:

The proposed model has a False Positive Rate (FPR) of 5.2% and a False Negative Rate (FNR) of 6.4%. The low false alarm rate suggests the system can be run continuously without overwhelming security personnel with false alarms. The ROC-AUC is of 0.98 which confirms the highly discriminative ability of the model across different decision thresholds.

5.2.6 Real-Time Edge Deployment Performance

Testing the system in a real-world scenario using a **Raspberry Pi 5** which has a compact, low-cost computer and confirmed its practical effectiveness. The AI processes will lost approximately **13 images per second**, which provides a consistent enough stream which is reliable to detect and track movements. In terms of energy efficiency, the entire setup, which will includes the camera, that consumes **less than 8 Watts**. This is very less power than a single household LED light bulb, making the system ideal for remote border areas that rely on solar power or batteries. Furthermore, the total delay from detecting an intruder to issuing an alert is **under 130 milliseconds**, which is roughly the speed of a human eye blink. Because border security does not require an instantaneous, millisecond-level responses will be very effective, and this speed is sufficient. These results will demonstrates that high-tech surveillance can be successfully on small run, and edge devices which was energy-efficient rather than requiring massive, expensive hardware.

5.2.7 Comparative Analysis with Existing Techniques

A comparative evaluation was performed with existing models which includes YOLO-X, YOLO-R, and SIFT+BPNN models. Table III will represents the accuracy comparison:

From Table III, it has been observed that SIFT+BPNN performed moderately because of its reliance on handcrafted features, but it was difficult to generalise to different surveillance conditions. YOLO-based techniques, which are mainly used for object detection, also performed with a low classification accuracy in the case of temporal anomaly modelling for infiltration detection. The proposed MobileNet-ConvLSTM architecture has better performance compared to all other techniques based on improvement in classification metrics, false alarm rates, and efficient edge deployment, making it a good model for border surveillance systems.

5.3 Analysis and Interpretation of Results

The results presented in Section 5.2 are mentioned in below across are four analytical dimensions. They are classification quality, edge performance, class-level behaviour, and architectural effectiveness.

5.3.1 Classification Quality Analysis

The model achieved an accuracy of 92.8% on the held-out test set with a standard deviation of only $\pm 0.6\%$ was achieved across five training runs, and confirming both strong performance and reliable reproducibility. The following points will interpret the significance of each metrics:

- **High Recall (93.6%) is the most important metric for this application:** A border surveillance system must be minimise missing's of infiltrations alert. The model correctly detects over 9 in every 10 genuine infiltration sequences. The 6.4% FNR is operationally acceptable given that human operators supplement automated detection.
- **Precision of 91.2% ensures sustainable operation:** Fewer than 1 in 11 alerts is a false positive. This prevents alert fatigue and maintains operator trust in the automated system over continuous, unattended deployment.
- **ROC-AUC which is of 0.98 provides operational flexibility:** The near-perfect AUC means that if an operational threat level has changed, the decision threshold which can be adjusted without substantial accuracy loss. The model's probability scores are well-calibrated across the entire operating range, as clearly visible in Fig. 5.5.

- The Balanced Accuracy which is of 92.1% confirms no class bias: The near-equality between standard accuracy (92.8%) and Balanced Accuracy (92.1%) would confirm that the class-weighted training strategy has successfully counter-acted with the UCF-Crime dataset's class imbalance. The model is not inflating accuracy by which it is favouring the majority class.
- **Low standard deviation confirms deploy ability:** The maximum standard deviation across all metrics is $\pm 0.7\%$ (Recall), meaning any of the five trained model instances would produce nearly identical operational performance. This reproducibility is essential for safety-critical deployment.

5.3.2 Training Dynamics Analysis

The accuracy and loss curves (Figs. 5.2 and 5.3) reveal an clear two-phase convergence pattern which is consistent with multi-stage transfer learning:

- **Phase 1 : Spatial Adaptation (Early Epochs):** Both training and validation accuracy would rise gradually as MobileNetV3 adapts its ImageNet-pretrained weights to the surveillance domain. The learning rate was kept small to preserve the general low-level feature detectors while shifting the higher-level features towards the new domain.
- **Phase 2 : Temporal Convergence (Mid to Late Epochs):** Accuracy rises more steeply as the ConvLSTM layer begins learning temporal motion patterns on top of the stabilised spatial features. The model's capacity to recognise spatio-temporal infiltration patterns across unseen video sequences has been increased significantly with later epochs.
- **Generalisation gap below 2%:** The training accuracy would consistently lead by less than 2% over validation accuracy. This narrow gap that is, visible throughout Figs. 5.2 and 5.3, which demonstrates the combination of Dropout (0.3), L2 regularisation (1×10^{-5}), and early stopping effectively that was prevented overfitting.
- **No divergence in loss curves:** The training and validation loss curves had been decreased smoothly and stabilise together, with no evidence of the sharp divergence that has characterized overfitting. Minor mid-epoch oscillations are expected behaviour in Adam-optimised networks.

5.3.3 ROC Curve Analysis

The ROC curve (Fig. 5.5) demonstrates several important properties of the proposed model:

- **AUC = 0.98:** The curve rises sharply from the origin and quickly reaches the top-left corner, confirming that the model achieves very high TPR at very low FPR values which is the ideal characteristic for a surveillance system.
- **Operational Point at FPR 5.2%, TPR 93.6%:** The red dot on the curve marks the chosen 0.5 decision threshold operating point. This point lies well above the random classifier diagonal (dashed blue line), confirming that the model is performing strongly at this threshold.
- **Threshold flexibility:** The steep initial rise means the threshold can be lowered to increase Recall further (at the cost of slightly more false alarms) or raised to reduce false alarms (at the cost of slightly more missed detections), giving operators meaningful operational flexibility.

5.3.4 Edge Performance Analysis

The Raspberry Pi 5 deployment results confirm the practical viability of the system for remote border surveillance:

- **TFLite quantisation is lossless in practice:** The accuracy degradation from full float32 TensorFlow model (~220 ms per clip) to TFLite quantised model (78 ms per clip) was less than 0.3%, while achieving a nearly 3× inference speedup. This will confirms model's ability of numerical precision reduction.
- **13 FPS is sufficient for border surveillance:** At 13 FPS, the sliding 16-frame window will covers an approximately 1.2 seconds of footage, as well as providing sufficient temporal patterns to differentiate infiltration behaviours (like crawling, walking towards perimeter) from background activity.
- **Below 8W enabled solar-powered deployment:** With system has power under 8 Watts, the surveillance unit was compatible to work with small solar panels (like 10–15W), as well as enabling independent operation in remote, off-grid border environments where there is on grid power infrastructure.
- **End-to-end latency below 130 ms:** The complete pipeline was from raw frame capture to alert generation via inference which would complete under 130 ms,

as well as by confirming real-time capability. This result validates the practical deploying ability of the MobileNet-ConvLSTM model on edge devices.

5.4 Validation of Objectives

This section was validated in order where each primary objective of the project and has obtained the results. Each objective was assessed with its target criterion and has an evidence from the experimental results.

5.4.1 Objective 1 : Develop a Spatiotemporal Infiltration Detection Model

- **Objective:** Build a hybrid MobileNet-ConvLSTM model that jointly learns spatial appearance and temporal motion patterns from surveillance video for binary infiltration classification.
- **Target:** Accuracy > 85% on the held-out test set.
- **Result:** Accuracy = 92.8% ($\pm 0.6\%$) across five independent training runs; F1-Score = 92.4%.
- **Validation:** ACHIEVED with an exceeding target by 7.8 percentage points. The use of MobileNetV3 for spatial extraction combined with ConvLSTM for temporal modelling directly achieves this objective.

5.4.2 Objective 2 : Real-Time Deployment on Low-Power Edge Hardware

- **Objective:** Deploy the model on a Raspberry Pi 5 single-board computer for real-time continuous border surveillance.
- **Target:** Inference time < 150 ms; total system power < 10W.
- **Result:** Inference time = 78 ms; end-to-end latency < 130 ms; system power < 8W.
- **Validation:** ACHIEVED — both targets comfortably met. TFLite quantisation reduced inference time from 220 ms to 78 ms ($\sim 3\times$ improvement). Power budget of <8W is 20% below the 10W target.

5.4.3 Objective 3 : Maintain Low False Alarm Rate

- **Objective:** The system must produce sufficiently few false alarms to be operationally sustainable in continuous, unattended deployment.

- **Target:** False Positive Rate $< 10\%$.
- **Result:** FPR = 5.2% (after temporal smoothing post-processing filter).
- **Validation:** ACHIEVED and has met with significant headroom. The temporal smoothing filter (minimum 3 consecutive positive clips) reduced FPR from $\sim 9\%$ to 5.2% at only a 0.2% cost in Recall. The low false alarm rates suggest the system can be run continuously without overwhelming security personnel with false alarms.

5.4.4 Objective 4 : High Detection Rate (Low Miss Rate)

- **Objective:** The system must reliably detect genuine infiltration events; missed intrusions are the primary safety failure mode.
- **Target:** Recall $> 90\%$.
- **Result:** Recall = 93.6%; FNR = 6.4%.
- **Validation:** ACHIEVED and has target exceeded by 3.6 percentage points. The class-weighted binary cross-entropy loss directly drove this result by penalising missed infiltrations more heavily than false alarms.

5.4.5 Objective 5 : Train on Publicly Available Adapted Datasets

- **Objective:** Use publicly available UCF-Crime and VisDrone-VID datasets adapted via binary label mapping for the infiltration detection task.
- **Target:** Model generalises to unseen test clips with no data leakage.
- **Result:** Video-level splitting enforced; training-validation accuracy gap $< 2\%$ throughout training.
- **Validation:** ACHIEVED and has the binary re-mapping of UCF-Crime combined with VisDrone-VID produced a coherent training corpus. The narrow generalisation gap confirms genuine learning of transferable spatio-temporal features.

5.4.6 Objective 6 : Outperform Existing Baseline Methods

- **Objective:** The proposed system must improve upon existing spatial-only and handcrafted-feature methods for the same task.
- **Target:** Accuracy $> \text{SIFT+BPNN}$ (87%).

- **Result:** Proposed model: 92.8% vs SIFT+BPNN: 87% (+5.8 pp); vs YOLO-R: 60% (+32.8 pp).
- **Validation:** ACHIEVED — the proposed MobileNet-ConvLSTM architecture performed better compared to other existing techniques based on improvement in classification metrics, false alarm rates, and efficient edge deployment.

```

=====
🕒 15:39:06
✅ DETECTED: Stealing
📷 Confidence: 0.8391
=====

📁 Saved → detections\Stealing\20260424_153906\current.jpg
📡 Sent to server → Status: 200

=====
🕒 15:39:10
✅ DETECTED: Stealing
📷 Confidence: 0.8040
=====

```

Figure 13: Output of the MobileNet-ConvLSTM while Detecting

5.4.7 Objective 7 : Alert Transmission to Central Server

- **Objective:** Detection events must be transmitted to a central server for storage and monitoring.
- **Target:** End-to-end alert delivery functional within acceptable latency.
- **Result:** HTTP POST alert pipeline validated; events logged to database with metadata within < 130 ms end-to-end latency.
- **Validation:** ACHIEVED and when an infiltration event is detected, the relevant data has been sent to the central server where it is kept in a database for future analysis of the event and monitoring.



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\vishw\OneDrive\Desktop\BORDERDETECTION> node index.js
File: 1777612944729-current.jpg
🚨 Detection received:
Label: Stealing
Confidence: 0.8259189128875732
File: 1777612988380-current.jpg
🚨 Detection received:
Label: Stealing
Confidence: 0.9761056303977966
File: 1777613003784-current.jpg
🚨 Detection received:
Label: Stealing
Confidence: 0.8322776556015015
File: 1777613070644-current.jpg
🚨 Detection received:
Label: Stealing
Confidence: 0.949474573135376
File: 1777613083167-current.jpg
```

Figure 14 : Alert Message Received from MobileNet-ConvLSTM Model to Central Server

5.5 Comparison with Expected & Existing Outcomes

5.5.1 Comparison with Existing Techniques

A comparative evaluation was performed with existing techniques including YOLO-X, YOLO-R, and SIFT+BPNN on the same infiltration detection task. The following analysis interprets why the performance gaps exist:

vs YOLO-X (58.0%) and YOLO-R (60.0%)

YOLO-based techniques are mainly used for object detection — they process each video frame in isolation and classify what is present spatially. They performed with low classification accuracy in the case of temporal anomaly modelling for infiltration detection because they are fundamentally unable to capture the movement of objects and changing patterns of behaviour over time. The 32–35 percentage point gap between YOLO methods and the proposed model is directly attributable to the ConvLSTM temporal module, which learns how a scene evolves across 16 consecutive frames rather than what exists in a single frame.

vs SIFT+BPNN (87.0%)

SIFT+BPNN performed moderately because of its reliance on handcrafted Scale-Invariant Feature Transform descriptors. While SIFT implicitly captures some local motion information through frame differencing, handcrafted features cannot adapt to

the diverse lighting conditions, camera angles, and subject appearances found in real-world border environments. The 5.8 percentage point improvement of the proposed model demonstrates the advantage of end-to-end learned spatiotemporal representations over manually engineered descriptor

5.5.2 Comparison with Expected Outcomes:

The following points compare the expected targets defined at the outset of the project against the results actually obtained:

- Accuracy: Target: >85% → Obtained: 92.8% and has been exceeded by 7.8 pp.
- Recall: Target: >90% → Obtained: 93.6% and has been exceeded by 3.6 pp.
- F1-Score: Target: >85% → Obtained: 92.4% and has been exceeded by 7.4 pp.
- False Positive Rate: Target: <10% → Obtained: 5.2% and has been exceeded (nearly half the target threshold).
- ROC-AUC: Target: >0.90 → Obtained: 0.98 and has been exceeded and has near-perfect discriminability.
- Inference Time: Target: <150 ms → Obtained: 78 ms and has been exceeded and has nearly 2× faster than required.
- End-to-End Latency: Target: <200 ms → Obtained: <130 ms which has been exceeded.
- System Power: Target: <10W → Obtained: <8W which has been exceeded as well as enables solar-powered remote deployment.
- Training-Validation Gap: Target: <5% → Obtained: <2% which has been exceeded and has strong generalisation confirmed.

5.5.3 Actual Outcomes of MobileNet-ConvLSTM:

From , Figure 14 , the results indicates that the model performs as strongly in overall, by achieving 92% of accuracy and has a high weighted F1-score of 0.92, which suggests that it is an effective general classification tasks, particularly when considering class imbalance.

A key observation is that the excellent performance on the "Normal Videos" class (F1-score = 0.98), has dominating the dataset with a large number of samples. This will significantly contribute to the high overall accuracy and weighted metrics. Similarly,

classes like Burglary, Explosion, Fighting, Shoplifting, and Stealing will also demonstrates a consistently high precision and recall, indicating that the model can reliably to detect these activities.

However, the results also reveal notable a weaknesses in certain classes. For example: Shooting has a perfect precision (1.00) whereas extremely low recall (0.12), but meaning the model detects a very few actual instances despite being correct when it does.

Road Accidents and Vandalism has shown a low F1-scores, which is indicating poor balance between precision and recall. Assault has low precision (0.39) but has a high recall (0.82), which is suggesting a frequent false positive.

--- Classification Report ---				
	precision	recall	f1-score	support
Abuse	0.53	0.89	0.67	19
Arrest	0.70	0.62	0.65	26
Arson	0.86	0.75	0.80	24
Assault	0.39	0.82	0.53	11
Burglary	0.91	0.84	0.88	38
Explosion	0.78	0.95	0.86	19
Fighting	0.86	0.79	0.83	24
NormalVideos	0.98	0.97	0.98	889
RoadAccidents	0.73	0.35	0.47	23
Shooting	1.00	0.12	0.22	8
Shoplifting	0.90	0.79	0.84	24
Stealing	0.78	0.95	0.86	42
Vandalism	0.37	0.50	0.42	14
Visdrone_Object	1.00	1.00	1.00	7
accuracy			0.92	1168
macro avg	0.77	0.74	0.71	1168
weighted avg	0.93	0.92	0.92	1168

Figure 15 : Classification Report of Proposed MobileNet-ConvLSTM

These variations will also highlight the impact of class imbalance and insufficient training samples for some categories. The macro average F1-score (0.71) notably has the lower than the weighted average, which is reinforcing that performs an uneven across classes.

5.5.4 Overall Assessment:

The proposed MobileNet-ConvLSTM framework demonstrates balanced performance with effective accuracy, inference capabilities, statistical viability, and real-time

capabilities along with all vital requirements for effective intelligent border surveillance systems. Specifically:

1. Every pre-defined project objective was met, and the majority were substantially exceeded.
2. The proposed model outperforms all three compared existing methods by significant margins on the same infiltration detection task.
3. The real-time capability of the framework, when implemented on the Raspberry Pi 5, validates its practicality in real-world border surveillance. Moreover, the power consumption of the framework is low, which is a major requirement when operating in a remote environment.
4. The low generalisation gap observed between training and validation stages, confirmed by the narrow curves in Figs. 5.2 and 5.3, demonstrates that the model has learned genuinely transferable spatio-temporal features rather than memorising the training data.
5. The balanced precision/recall profile and low false alarm rates, supported by an ROC-AUC of 0.98 (Fig. 5.5), confirm the system's suitability for continuous, unattended border surveillance operations.

CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusion

This article proposes a lightweight spatio-temporal border surveillance framework using the MobileNet ConvLSTM architecture for effective infiltration detection. The proposed framework utilizes the benefits of spatial feature extraction using the MobileNet architecture and temporal sequence learning using the ConvLSTM architecture. This framework is effective in extracting motion-dependent behavioral patterns rather than spatial features. The proposed framework is trained using a multi-stage training approach. As a result, the framework is effective in generalizing the training using the VisDrone-VID dataset for spatial adaptation and the UCF Crime dataset for temporal infiltration detection.

The experimental analysis performed on the proposed framework shows the effective convergence behavior of the proposed framework. The low generalization gap is observed between the training and validation stages. The statistical analysis performed on the proposed framework shows the consistent performance of the proposed framework with low variance in the results of various experiments performed using the framework. The proposed framework is effective in ensuring the accuracy of classification results with balanced precision/recall rates and low false alarm rates using the ROC curve analysis. The experimental analysis performed on the proposed framework shows the effective detection capabilities of the proposed framework for infiltration detection with low false alarm rates, a vital requirement for effective border surveillance systems.

In addition, the real-time capability of the proposed framework, when implemented on the Raspberry Pi 5, validates its practicality in real-world border surveillance. Moreover, the power consumption of the proposed framework is low, which is a major requirement when operating in a remote environment. The proposed MobileNet ConvLSTM framework is effective in demonstrating balanced performance with effective accuracy, inference capabilities, statistical viability, and real-time capabilities, which are vital requirement for effective intelligent border surveillance systems.

6.2 Future Scope

The proposed work can be extended in multiple directions to which might enhance its effectiveness and applicability's. One potential improvement is that to move beyond binary classification and develop a multi-class framework that can identify different types of intrusion activities, such as armed threats, group movements, or suspicious behaviours. Incorporating advanced architectures such as attention mechanisms or transformer-based models will also improve the system's scope which able to capture long-term temporal dependencies and contextual relationships. The framework can also be expanded to support multi-camera and distributed edge surveillance systems for large-scale border monitoring. To enhance model under adverse conditions of model by integrating methods like thermal imaging, infrared sensors, or image enhancing techniques would have improved performance of model in low-visibility environments. Additionally, by enabling adaptive or online learning would also allow the model to continuously update itself based on new data, that would have make more resilient to evolving intrusion patterns over future. Furthermore, the speed and latency will be improved using optimization techniques like model pruning, quantization, and specialized edge accelerators. Finally, by integrating the system into a comprehensive surveillance ecosystem which has an automated alert systems, geographic mapping, and multimodal data fusion (e.g., audio or radar inputs) will improve the practicality and reliability of the solution for real-world applications.

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APPENDICES

APPENDIX A: CONTRIBUTION OF TEAM MEMBERS

The following table outlines the individual contributions of each team member toward the successful completion of the major project titled "**MobileNet–ConvLSTM Based Low-Power Spatio-Temporal Surveillance for Border Security.**"

S.No	Team Member	Roll No	Contribution
1	A. Anshul	22071A6767	Dataset collection, Literature review support and presentation preparation
2	K .Vishweshwar Reddy	22071A6784	Dataset collection, preprocessing pipeline design, binary label mapping, data partitioning, MobileNetV3 backbone integration, model training, documentation, and patent filing support
3	K. Hitesh Varma	22071A6785	ConvLSTM layer design, temporal sequence modeling, hyperparameter tuning, model evaluation, result documentation, report preparation and patent filing support.
4	M. Ganesh	22071A6798	ROC curve generation, performance evaluation, comparative analysis, confusion matrix analysis, Website Building

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